



Piezoelectric polarization-induced photogenerated carriers transfer coupled with active electron transport Z-scheme heterojunction synergistic optimization of BiFeO₃/CdS composite photocatalysts for hydrogen production

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ARTICLE INFO

Keywords:

Photocatalytic hydrogen generation
Z-scheme heterojunction
Carriers spatial separation
Charges transfer
Photo-piezoelectric coupling effect

ABSTRACT

Photocatalytic hydrogen production technology is an effective way to convert solar energy into chemical energy with the advantage of cleanliness and non-pollution, whereas efficient, stable and inexpensive photocatalysts are the top priority. The efficiency of separation, transportation and utilisation of photogenerated carriers is fundamental in limiting the rate of photocatalytic hydrogen production reactions. In this study, Z-scheme heterostructures with active electron transport properties are constructed with multiferroic BiFeO₃ and favorable visible light-responsive CdS nanosheets. The active electron transport system is constructed to promote carrier separation and transport, which presents a novel strategy for the enhancement of photocatalytic hydrogen production reaction activity. The heterostructure of the composites not only optimises the light absorption capacity and band structure, but also reduces the interfacial resistance and photogenerated carrier recombination. The combination of BiFeO₃ particles and ultrathin CdS not only suppresses the agglomeration of CdS nanosheets, but also assists in enhancing the specific surface area of the reaction system. More importantly, the nanosheet structure of CdS as the reducing phase facilitates proton adsorption and hydrogen desorption. The piezoelectric effect possessed by BiFeO₃ and CdS leads to the triggering of spontaneous polarisation moments in response to external forces. The introduction of polarisation factors can enhance the driving force of charge separation in the bulk phase thereby forming an internal electric field. Photogenerated electron-hole pairs during photocatalysis migrate to the surfaces of different polar domains under the influence of the internal electric field. The piezoelectric polarisation process exhibits significantly enhanced properties for the charge redistribution in the bulk phase of the catalyst as well as inducing the efficient separation of photogenerated carriers with further migration at the interface through the Z-mechanism heterojunction. The hydrogen evolution rate of the composite containing 20 wt% CdS reaches 3.58 mmol·g⁻¹·h⁻¹ under photo-piezoelectric coupling conditions, which is 116.64 fold higher than that of the pristine BiFeO₃ under visible light irradiation. This study will provide an important theoretical basis and practical guidance on the application of multifield coupling to modulate charge transfer as well as inducing interfacial catalytic reactions.

1. Introduction

Semiconductor photocatalysis has emerged as promising technology

for solar energy conversion because it is able to realize energy conversion by sunlight or artificial light sources, and is therefore widely used around the world. So far, although attempts have been devoted to the

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<https://doi.org/10.1016/j.jece.2024.115049>

Received 23 October 2024; Received in revised form 15 November 2024; Accepted 5 December 2024

Available online 12 December 2024

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fabrication of different visible light driven photocatalysts, there are still problems such as the short lifetime of photogenerated carriers and the limited absorption of visible light, which limit the photocatalysis efficiencies. About 90 % of the photogenerated carriers generated in photocatalytic materials after photoexcitation are rapidly complexed [11]. This matter seriously limits the industrialisation of photocatalytic technology. Therefore, the most urgent task to improve the catalytic property of photocatalysts is to improve the separation efficiency of photogenerated carriers. For the improvement of the photogenerated carrier separation process, it is necessary to deeply analyze the principle of photocatalytic materials in the process of solar energy action, so as to design an efficient photocatalytic solar energy conversion system, improve the photocatalytic energy conversion efficiency, and strive to realize the industrial scale development of photocatalytic technology. In view of the importance of the development of hydrogen energy in alleviating the current energy crisis and environmental pollution, it is necessary to develop more forms of energy and to combine other forms of energy with solar energy for hydrogen production from water splitting while utilising solar energy for the photocatalytic [2,3]. Therefore, it is necessary to combine the specific structure and properties of semiconductors to modulate them so that they exhibit outstanding hydrogen production performance under the excitation of different forms of energy. Recently, research on modulating the reaction activity of semiconductor photocatalysis with the help of external fields has been emerging. In photocatalytic systems with external field response performance, the auxiliary microwave [4], temperature field [5], or external electric field [6] all contribute to suppress the photogenerated carriers recombination and promote the reverse migration of photogenerated electrons and holes in semiconductor photocatalytic materials. Especially, polarization field engineering is gradually becoming one of the most effective strategies to augment the efficiency of photogenerated carriers between the bulk phase and the surface separation.

Polarity is widely found in non-centrosymmetric materials such as piezoelectric, thermoelectric, ferroelectric and nonlinear optical materials. The non-centrosymmetric arrangement of atoms or atoms groups in the crystal structure of a non-centrosymmetric material (non-overlapping positive and negative charge centres) results in the arrangement of positive and negative charges on opposite sides of the crystal structure, which generates a polarised electric field pointing from the negative charge region to the positive charge region [7]. The emergence of polarized electric field in photocatalysis can effectively promote the photogenerated carriers separation [8]. Polarised electric fields function as a direct actuating factor in some photoresponsive and piezoresponsive catalysts, where the polarisation electric field is able to drive photogenerated electrons (e^-) and holes (h^+) towards positively and negatively charged regions, respectively, due to the charge shielding effect. This process allows for the rapid transport of carriers from the bulk phase to the opposite surface during photocatalysis, enabling the spatial separation of redox sites [9]. Meanwhile, the polarized electric field accelerates the surface charge transfer of the photocatalyst, further improving the catalytic efficiency.

When a certain wavelength range of light is absorbed by the catalyst, the semiconductor catalyst is excited by light and the electrons migrate from the valence band (VB) to the conduction band (CB) of the catalyst, forming photogenerated electron-hole pairs. The photogenerated carriers will migrate out of the bulk phase towards the catalytically active sites on the catalyst surface and participate in the redox reaction, with the carrier mobilisation process typically taking several hundred picoseconds. Nevertheless, photogenerated carriers recombination in the bulk phase takes only a few picoseconds, which is one percent of carrier migration [10].

Therefore, there is an immediate need to use the polarisation field as a driving force to promote carriers separation in the bulk phase and thus improve the photocatalytic performance. Huang et al. have linked the introduction of transition metal cations to the deformation of SiO_4 tetrahedra and the generation of polarization fields through the

accumulation of SiO_4 tetrahedra in $\text{Ag}_6\text{Si}_2\text{O}_7$ [11]. Theoretical calculations show that the photogenerated electron migration from $\text{Ag}(\text{c})\text{O}_4$ to $\text{Ag}(\text{b})\text{O}_3$ and then to $\text{Ag}(\text{a})\text{O}_2$ is consistent with the direction of the internally polarized electric field generated by the polar SiO_4 tetrahedral arrangement. Ding et al. surface-modified polymerized carbon nitride ($\text{g-C}_3\text{N}_4$) with a graphite carbon ring to establish an internal electric field that caused light-induced e^- and h^+ to move in different directions, thereby increasing the photocatalytic hydrogen production rate by 21 times [12]. Furthermore, bismuth-based perovskite oxides with a narrow bandgap and good ferroelectric response are suitable for photogenerated carrier separation [13,14]. Liu et al. used the piezoelectric polarization of BiFeO_3 in a core-shell structure to confine the photogenerated electrons and holes of TiO_2 to the positive and negative sides of the piezoelectric, respectively [15]. Due to the fact that there is no centre of symmetry in the crystalline structure for polar materials, the accumulated microscopic polarisation in the polar unit of a piezoelectric photocatalyst produces a large macroscopic polarisation when the catalyst is subjected to mechanical stresses. The positive and negative charge centers of the ions inside the crystals are displaced relative to each other, ion polarization occurs, and a piezoelectric potential is generated [16], which in turn disrupts the original charge equilibrium state on the surface of the material [17,18]. Subsequently, when the material surface is in a stable state with a sufficiently large Gibbs free energy and effectively contacts with reactants (water, organic small molecules, gas molecules), redox reactions at the surface interface will be triggered, resulting in the loss and gain of electrons, which converts mechanical energy into chemical energy [19–23].

It is noteworthy that improving the utilization of photogenerated carriers in the chemical reaction is another essential to enhance the photocatalytic performance. The construction of band-matched heterostructured photocatalysts through the differences in the band structures of different semiconductors can utilize the potential energy difference to form a spontaneous internal electric field at the composite interface, realizing the spontaneous movement of photogenerated carriers from the high-energy level to the low-energy level, which in turn serves as another effective strategy to improve the carrier separation efficiency of photocatalysts [24]. Heterojunction composites composed of semiconductor-semiconductor (S1-S2) are usually most common as type-I and type-II heterojunctions. Type-I heterojunctions consist of two- or multi-component semiconductors in which S2 has a higher CB than S1 and S2 has a lower VB than S1, thus holes and electrons are transferred and accumulate on S1. The photogenerated electron transfer from S2 to S1 in type-II heterojunctions depends on the CB of S2 in a more negative position. On the contrary, holes can migrate from S1 with more positive VB towards S2, resulting in spatial carriers separation and enhanced photocatalytic reactivity. Although the type-II heterojunction performs perfectly in charge separation, there are still some shortcomings.

Because the transfer of electrons from higher to lower energy level and holes from lower to higher energy level makes the reacting system have a more positive CB and a more negative VB, in thermodynamics this charge transfer mechanism is detrimental to the redox capacity of the reacting system. In addition, on the kinetic side, the repulsive forces between electrons and electrons (holes and holes) inhibit the hypothesised charge transfer mode. In terms of energy utilisation, the CB or VB energy difference between S2 and S1 is not involved in the photocatalytic reaction. In relation to the decreasing redox capacity of type-II heterojunctions, Z-scheme heterojunctions have a more suitable charge transfer system for redox reactions. Photogenerated electrons in semiconductor S1 having a low reduction capacity are compounded with photogenerated holes in semiconductor S2 possessing reduced oxidation capacity. Consequently, it is possible to maintain photogenerated electrons in the semiconductor S2 possessing a high reduction capability and photogenerated holes in the semiconductor S1 with a high oxidation capability. The redox capability of the Z-scheme photocatalysts has been optimised. It is also important to note that carrier mobilisation for Z-scheme photocatalysts is physiologically more viable than that for type-

II heterojunction photocatalysts, since the photogenerated electrons migrating from the CB of the semiconductor S1 to the photogenerated holes VB of the semiconductor S2 are advantageous due to the electrostatic attraction between the electrons and the holes.

Some researches have shown that the heterogeneous structure constructed in the CdS supported Z-scheme system helps to avoid the reverse reaction involving redox mediators, while the polar surface of the polar catalyst plays a key role in mediating the interfacial electron transfer in the CdS-based composite system [25–27]. A direct Z-scheme heterojunction formation between the CdS/ WO_3 composites prepared by Jin et al. improves carrier separation efficiency, which in turn exhibited a better photocatalytic methane production efficiency [28]. Yu et al. constructed an interfacial linker coordinated heterostructured cadmium sulfide/zirconium-based metal-organic framework (CdS/NU-M) to realize a Z-scheme charge transfer channel in CdS/NU-M. As the photogenerated charges were efficiently separated and transported to the surface, it induced a 10 times higher photocatalytic hydrogen production rate of CdS/NU-M than that of the pure phase photocatalyst [29]. Wu et al. constructed a Z-scheme $\text{BiFeO}_3/\text{Bi}_2\text{O}_4$ heterostructure by anchoring BiFeO_3 nanoparticles to Bi_2O_4 with a rod structure, and elucidated the charge transfer pathway between BiFeO_3 and Bi_2O_4 based on the band structure and the main active substances [30].

Therefore, the typical piezoelectric photocatalysts CdS and BiFeO_3 are used as the research objects in this paper, and the Z-scheme heterojunction composites of BiFeO_3 coated with ultrathin CdS are successfully prepared. The performance of dual internal electric fields on water splitting and the mechanism for photo-piezoelectric coupling action on photogenerated carrier transport are investigated.

2. Experimental

2.1. Materials preparation

BiFeO_3 (BFO) powders preparation:

The chemical reagents used in this work were all analytical grade reagents purchased from the Sinopharm Chemical Reagent Co., Ltd. First of all, 0.011 mol $\text{Bi}(\text{NO}_3)_3$ and 0.01 mol $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ were weighed and dissolved into 100 mL of deionized water, respectively, to obtain an aqueous solution of 0.11 M bismuth nitrate and 0.1 M iron nitrate. Wherein the solvent of bismuth nitrate was added 3 mL of nitric acid (68 %) gradually during constant stirring until the bismuth nitrate was completely dissolved. Then the two solutions were mixed and 0.02 mol of triammonium citrate was added as the complexing agent and stirred for 30 min, and the gel of the reaction intermediate was obtained by continuous stirring at 90°C until the water was completely evaporated. The gel was calcined at 550°C for 2 h to obtain pure phase BFO powder particles. The preparation process of BFO samples is shown in Fig. 1a.

CdS nanosheets preparation:

After adding 10 mL of 0.3 M cadmium chloride solution to 50 mL of 0.1 M sodium citrate solution and stirring at room temperature for 20 min. Then, adding 3 mL of concentrated ammonia ($\text{NH}_3 \cdot \text{H}_2\text{O}$ 28 %) and 10 mL of 0.6 M thiourea solution in sequence. The mixed solution was then reacted in a water bath at 60°C for 2.5 h. After being cooled naturally, the precipitate was collected by centrifugation, rinsed several times with deionized water and anhydrous ethanol to remove impurities, and dried in an oven at 60°C to obtain CdS nanosheets. The preparation process of CdS samples is shown in Fig. 1b.

BFO/CdS samples preparation:

0.6 g of BFO was weighed into 60 mL of 0.1 M sodium citrate solution, and 1.38 mL of 0.3 M cadmium chloride solution was added according to 10 wt% of BFO, and stirred at room temperature for 3 h. Then 3.6 mL of concentrated ammonia ($\text{NH}_3 \cdot \text{H}_2\text{O}$, 28 %) and 1.38 mL of 0.9 M thiourea solution were slowly added; the reaction was then carried out at 60°C in a water bath for 2.5 h. The precipitates were centrifuged after natural cooling. The precipitate was collected, and the precipitate was rinsed several times with deionised water and anhydrous ethanol, and dried in an oven at 60°C to obtain 10 % CdS/ BiFeO_3 (BC10) samples. Cadmium chloride solution and thiourea solution could be further increased in stoichiometric ratios of the corresponding volume

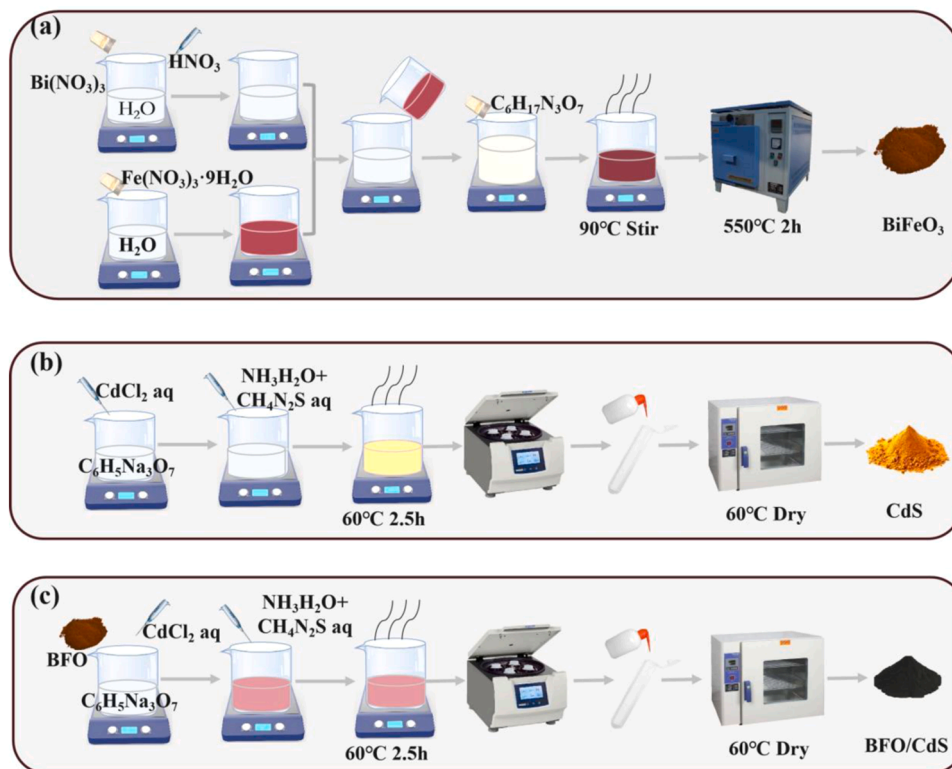


Fig. 1. The preparation process of (a) BiFeO_3 powders, (b) CdS nanosheets, (c) BFO/CdS samples.

solutions to obtain 20 % CdS/BiFeO₃ (BC20) and 30 % CdS/BiFeO₃ (BC30). And the preparation process of BFO/CdS samples is displayed in Fig. 1c.

2.2. Catalysts characterization

The crystal structure and phase composition of the powders were determined using a Smart Lab X-ray diffraction (XRD) of Rigaku Corporation (Cu target, voltage: 40 kV, current: 40 mA, scanning range: 2 θ =20–80). The surface morphology of the samples was observed using a SU8020 scanning electron microscope (SEM) of Hitachi (HITACHI). Meanwhile, the electron migration direction and lattice spacing of the samples were determined by FEI Tecnai G2F30 S-Twin transmission electron microscope (TEM). The chemical compositions and states of the sample surfaces were analyzed using a Kratos' Axis Ultra X-ray photoelectron spectrometer (XPS). The low-temperature adsorption-desorption curves of the powders under N₂ atmosphere were tested using ASAP 2020 from Micromeritics and the specific surface area and pore structure of the samples were determined. The photoluminescence properties of the samples were analyzed using HORIBA's Fluorolog-QM steady-state transient fluorescence phosphorescence spectrometer (PTI) with a Xenon lamp source at 420 nm to characterize the separation and recombination of photogenerated electron-hole pairs as well as the defect concentration of the photocatalysts. Electron paramagnetic resonance spectroscopy (EPR) was characterized by an electron paramagnetic resonance spectrometer (Bruker ER075), which determines the state of the unpaired electrons by monitoring the absorption signals of the electrons associated with the magnetic field. UV-Vis absorption spectra were obtained by a Cary 5000 UV-Vis-nir spectrophotometer (Agilent Technologies) to analyze the band gap and light absorption properties of the catalyst. The band gap of the sample was estimated according to the Tauc equation: $\alpha h\nu = A(h\nu - E_g)^{n/2}$, where α is the absorption coefficient, h is the Planck's constant, ν is the optical frequency, A is the proportionality constant and E_g is the bandgap of the sample (n is taken to be 1 for indirect bandgap semiconductors and 4 for direct bandgap semiconductors).

2.3. Photoelectrochemical tests

Preparation of film samples for partial photoelectrochemical testing: 0.010 g of sample and 5 mL of rosinol were added to a centrifuge tube and the mixture was treated under ultrasound for 120 min to form a homogeneous suspension. 200 μ L of the suspension was added dropwise to an area of 1.5 cm² (1.5 $\times$ 1.0 cm²) on indium tin oxide (ITO) glass and stainless steel sheets and dried in an oven at 240°C. The suspension was then dried in an oven at 240°C. The BFO, BC10, BC20, BC30 and CdS films were then successfully prepared on ITO glass and stainless steel sheets, respectively.

The photoelectric performance of the catalysts was carried out in a three-electrode system on a CHI 660E electrochemical workstation, with an FTO glass catalyst as the working electrode, a platinum plate as the counter electrode, and Ag/AgCl as the reference electrode. An external 300 W Xenon lamp was used as the light source. The electrolyte was 0.1 M Na₂SO₄ solution. The transient current density of the catalyst at rated voltage was recorded to characterize the photoelectric response of the catalyst. The relationship between the potential between the sample and electrolyte interface and the space charge layer capacitance was characterized by the Mott-Schottky test in a standard three-electrode system.

2.4. Catalytic performance test

Measure the photocatalytic, piezoelectric, and photocatalytic activities of the sample by photocatalytic decomposition of water to produce hydrogen. Select ultrasonic wave source (100 W, 40 K Hz, GL0203–40, China) and Xenon lamp (300 W, CEL-HXF300, China) with the UV cutoff

filter ($\lambda \geq 420$ nm) to provide periodic mechanical load and simulated solar light source, respectively. In the test, 20 mg of catalyst powder samples were weighed and dissolved in 100 mL of 0.35 M Na₂S/0.25 M Na₂SO₃ mixed sacrificial solution. Before the reaction, a vacuum pump was used to evacuate for 30 min to remove air from the reaction system. Then, during light source irradiation and/or ultrasonic treatment, analyze the hydrogen gas produced by photocatalytic water splitting using gas chromatography (GC-7920).

2.5. Piezoelectric performance test

Piezoelectric coefficient test: The upper and lower surfaces of different thin film samples on stainless steel sheets are coated with high-temperature silver paste using screen printing method, and the films are heated to 500 °C at a rate of 3 °C/min for 2 hours with silver backing. After covering a layer of copper sheet, the YE2730A quasi-static d₃₃ measuring instrument is used for d₃₃ testing;

Current voltage measurement (I-V): ITO glass was covered on the surface of a thin film sample and placed in the dark. During the measurement, a periodic force (10 N) was applied perpendicular to the thin film sample to compress the film. The I-V curve was recorded using a Keithley 2601 A digital source meter measuring device connected to two ITO layers.

3. Results and discussion

3.1. Composition analysis and crystalline structure

The hexagonal BFO has a typical perovskite crystal structure, which not only has a typical piezoelectric response but also possesses a favorable photoresponse. CdS as a typical metal sulfide possesses not only visible light response but also can be optimized to improve the charge transport path through the preparation process. The contact interface of the two-phase catalyst samples is more favorable for the electron transport through the interface of the catalysts, which allows for the desorption to generate hydrogen. The XRD pattern of BFO, ultrathin CdS nanosheets and BC10, BC20, BC30 composite samples prepared by the liquid-phase method as can be seen in Fig. 2a. The XRD patterns of composite samples presented characteristic diffraction peaks of BFO (JCPDS #72–2035) and CdS (JCPDF #89–2944). It is possible to find that the diffraction peaks of all BFO samples in both pristine samples and composite samples are sharp and intense, indicating that the prepared BFO catalysts are highly crystalline. Besides, no impurity phases such as Bi₂O₃ and Fe₃O₄ are generated. The BFO-containing samples generate the intensest characteristic peak at 32.1°, which corresponds to the (110) crystal plane of the hexagonal crystal system for BFO. Meanwhile the CdS nanosheets prepared by liquid-phase precipitation method not only have a wider half-peak width at the same time (002) crystalline planes present at 26.5° are much more selectively oriented than the other crystalline planes, which may play a pivotal effect on the carrier transport rate of the composite samples. The XRD diffraction peaks of each sample did not show impurity peaks, indicating that the composite photocatalysts with different ratios have been successfully prepared and the two components are well compatible.

The chemical composition and chemical state on the surface of each catalyst sample are then further characterised using XPS testing techniques to demonstrate the influence of the strong interaction between BFO and CdS and the construction of heterojunctions on electron transportation. The full spectra of BFO, CdS and BC20 composite catalysts are shown in Fig. 2(b), the elements Bi, Fe and O are present in BFO, the elements Cd and S are present in CdS, and BC20 samples contain all these five elements at the same time. In order to determine whether the binding energies of the elements in the heterojunction materials change or not, the elemental binding energies of the pure phase samples and the composite samples were analysed separately. The Bi 4f peaks located at 158.4 eV and 163.7 eV in the XPS spectrum of BFO originate from Bi

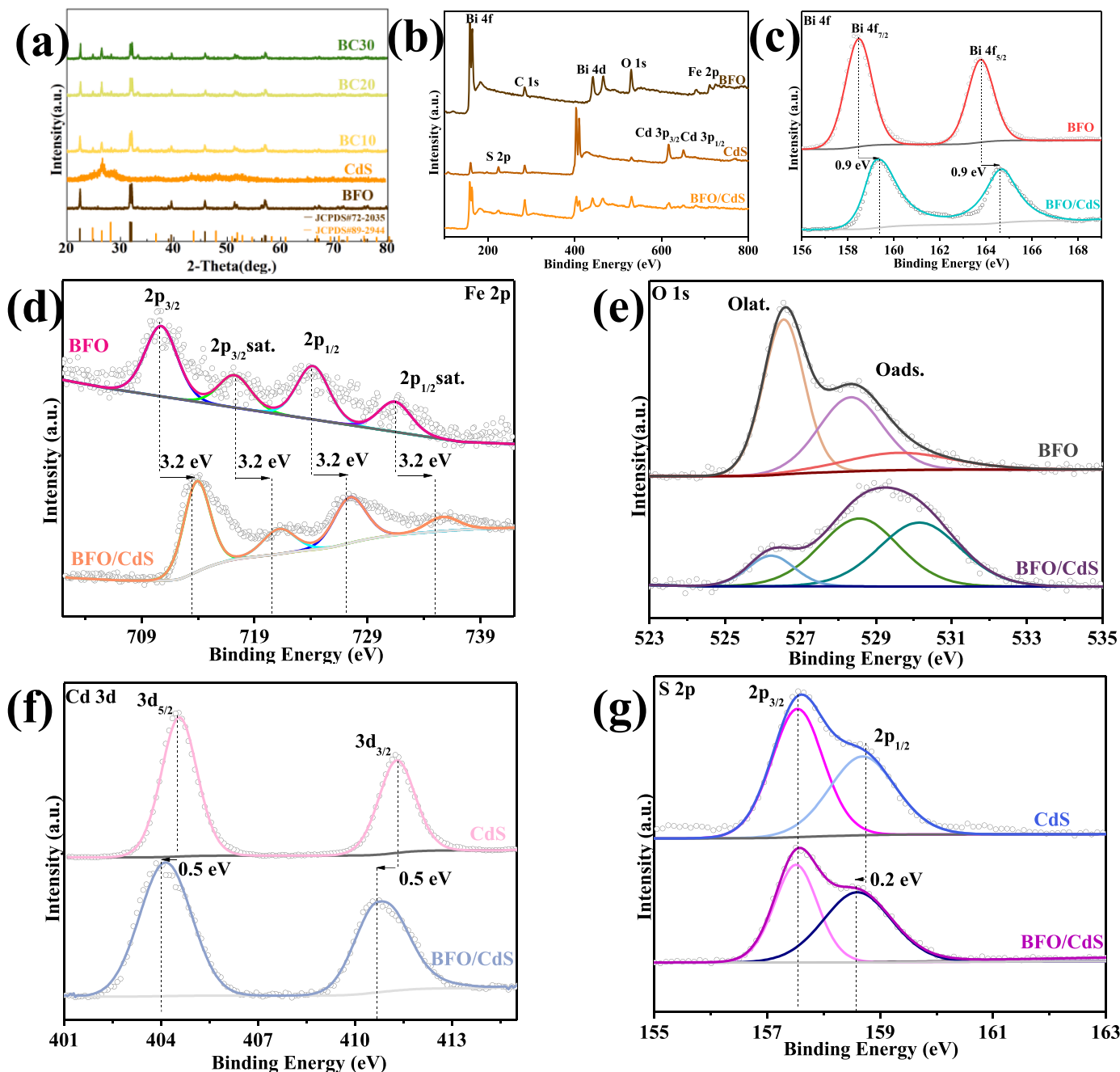


Fig. 2. (a) The XRD patterns of BFO powders, CdS nanosheets, BC10, BC20 and BC30 samples, (b) XPS spectra and high-resolution XPS spectra of (c) Bi 4 f, (d) Fe 2p, (e) O 1 s, (f) Cd 3d and (g) S 2p for the pristine BFO, CdS and BFO/BFO heterojunction composites.

4 $f_{7/2}$ and Bi 4 $f_{5/2}$, respectively [31], and these two binding energy peak positions are consistent with those of BFO in the full spectrum as shown in Fig. 2c. However, these two peak positions corresponding to BC20 are shifted to 159.3 eV and 164.6 eV, respectively. There are two characteristic peaks in the Fe 2p XPS pattern of BFO with binding energies of 724.1 eV and 710.7 eV, which originate from the binding energies of Fe 2p $_{1/2}$ and Fe 2p $_{3/2}$, respectively [32]. There is a satellite peak (~717.3 eV) at 4.8 eV higher than the binding energy of Fe 2p $_{3/2}$, which indicates that the Fe element in the material exists in the form of Fe³⁺ ions rather than Fe²⁺ ions [33]. The binding energy of Fe element in the composite sample is shifted to higher energy levels by 3.2 eV similarly to that of the Bi element. This confirms the existence of a potential charge transfer tendency between the two components in the heterojunction. It is noteworthy that the O 1s characteristic peaks of BC20 composites

correspond to lattice oxygen at low energy level, reactive oxygen at intermediate energy level and adsorbed oxygen at the high energy level, respectively. Where O_{ads}/O_{lat} reflects the surface oxygen reduction capacity of the catalyst [34–37]. The adsorbed oxygen not only has a large increase in peak area, but also tends to lose electrons by shifting to higher energy levels. This indicates that the construction of heterojunction between BFO and CdS effectively enhanced the surface redox reaction ability of the catalyst. Protons in the reaction medium are selectively attracted by the adsorbed oxygen through electrostatic gravity and thermal motion, which compensates for the negative effect of the lack of electron transport efficiency due to the rapid recombination of carriers. The high-resolution XPS spectra of Cd 3d of CdS and BC20 samples consist with two characteristic peaks of Cd 3d $_{5/2}$ and Cd 3d $_{3/2}$ locate at 404.4 and 411.2 eV, respectively. The characteristic

peaks of Cd 3d in the formed heterojunction composites are shifted to the lower energy level by 0.5 eV with a tendency to gain electrons. In contrast, the change in S 2p binding energy is not very pronounced, suggesting that the electron mobility in the heterojunction composites is more favorable from BFO to the metal cation around Cd^{2+} .

The individual catalyst samples were characterised and analysed by SEM to investigate the size, morphology and contact method of the prepared samples. As can be seen in Fig. 3a, the synthesised BFO particles are distributed in the range of 100–300 nm with the smooth surface of the material having spindle and ellipsoidal particle shapes. The particles are interlaced with each other to form a lot of pore-like structures, which is favourable to increase the contact area between the materials and the other phase catalysts together with the reaction solution, to effectively improve the adsorption effect. The morphology and structure of the pristine CdS were also analysed as shown in Fig. 3b. CdS is a nanoflower structure assembled from two-dimensional nanosheets and the thickness of individual nanosheets is in the nanoscale. The multiple nanosheets curl to form nanoflowers greatly increase the contact area with the reaction solution, and the ultrathin lamellar structure also helps to shorten the charge migration path from the bulk phase to the active sites on the catalyst surface. The SEM images of the BFO/CdS composite samples with CdS nanosheets gradually encapsulate on the surface of BFO particles are shown in Fig. 3c–e. With the gradual

increase in the mass fraction of CdS, BC20 is more fully distributed on the BFO surface due to the elevated percentage of nanoflowering CdS. The particulate BFO as a carrier of CdS enabled BC20 not only to have a higher contact area than BC10, but also inhibited the agglomeration of CdS, thus showing better dispersion. This offers the possibility of expanding the specific surface area of the composite samples and thus generating more reactive active sites. However, when the mass fraction of CdS is too high, the CdS nanosheets in the surface layer of BC30 have certain agglomeration phenomena and are no longer as unfolded as that of BC20, which may adversely affect the mass transfer of the reaction system. Subsequently, in order to demonstrate the successful preparation and dispersion of the composite BFO/CdS samples, EDS mapping of the elemental distribution in the BC20 samples was carried out. It can be seen that the distribution profiles of Bi and Fe elements are consistent with the sample morphology in Fig. 3f, which is due to the fact that the main structure of BC20 is composed of BFO. The distribution of Cd and S elements does not have the same profile as that of Bi or Fe elements and is more uniformly dispersed, which is attributed to the high dispersion of CdS and the coating structure of the composites. Further, the atomic percentage matches with the synthesised composite catalysts not only confirms the successful preparation of the samples in XRD, but also indicates the good dispersion of the samples.

In order to investigate the morphology characteristics and composite

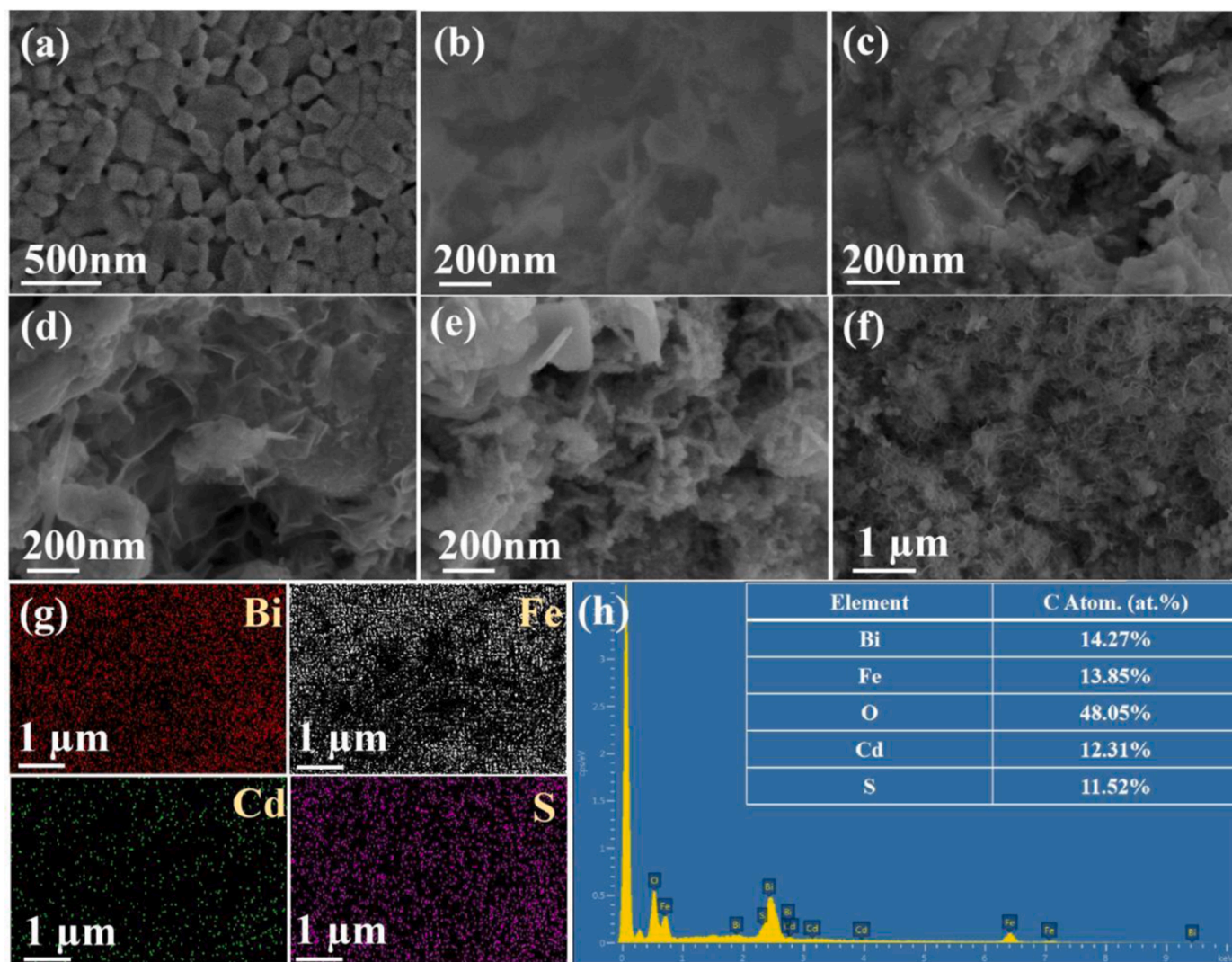


Fig. 3. SEM images of the (a) pristine BFO, (b) pristine CdS and (c) BC10, (d) BC20, (e) BC30 heterojunction composites. (f) The SEM images of BC20, (g) the corresponding area element mapping images with element Bi, Fe, Cd, and S. (h) The atomic percentage of corresponding area BC20 samples.

patterns of the heterojunction composite catalysts, BC20 was observed by transmission electron microscopy (TEM). Besides, to investigate the electron flow between the two components, 1 wt% of Pt particles were loaded on the sample surface by photo deposition through $\text{H}_2\text{PtCl}_6 \cdot 6\text{H}_2\text{O}$ precursor. Fig. 4a shows the obvious two-phase interface connection between the BFO of the bulk structure and the lamellar CdS. On the basis of HRTEM in Fig. 4b, it is observed that Pt microparticles are selectively deployed on the edge active sites of CdS nanosheets, which confirms that the electrons between the two components are transferred from BFO to CdS and reduced by the active sites of CdS. According to HRTEM Fig. 4b, it can be found that Pt nanoparticles are selectively deposited on the edge active sites of CdS nanosheets, which confirms that the electrons between the two components are transferred from BFO to CdS and reduced by the active sites of CdS. Furthermore, the tight binding of the contact interface with the successful preparation of the heterojunction is further supported by the lattice fringes of the two components in Fig. 4c. It is further found that the CdS nanosheets are preferentially oriented to the (002) lattice plane by Fig. 4d, which in agreement with the results in the XRD spectra.

For the purpose of determining the effect of the introduction of non-homogeneous structured CdS nanosheets on the pore structure of the composites, N_2 adsorption-desorption curves were employed to characterise the variation in the pore structure of the pristine BFO, CdS, and BFO/CdS composite photocatalysts with different mass ratios. The isothermal N_2 adsorption-desorption curves of each catalyst are shown in Fig. 5a. The horizontal coordinate is the relative pressure of nitrogen with respect to its saturation vapour pressure, and the vertical coordinate is the adsorbed amount of nitrogen. All the samples exhibit Langmuir type IV isotherms except BFO crystals, which contain H3-type hysteresis lines. Based on the classification criteria of the International Union of Pure and Applied Chemistry (IUPAC) [38], which indicates that the material contains a high number of micropores in addition to mesopores. The incident light is reflected and scattered several times through the mesoporous structure, which in turn enhances the absorption of the incident light. The microporous structure favours proton adsorption in solution and accelerated hydrogen desorption rate. The pore size distribution in Fig. 5b also confirms this view, and when the mass fraction of CdS reaches 20 %, the porosity of the sample is not only much higher than that of the pristine BFO but also the highest among all composite samples. The reaction medium in solution is affected by the adsorption of high specific surface area CdS coating on the surface of BFO, which transforms the original free motion in three-dimensional

space into restricted two-dimensional planar motion. The entropy decreases with decreasing disorder of the reaction system, which favors the rate of hydrogen reduction reaction on the catalyst surface. Also noteworthy is the porosity of the BC30 sample is rather unsatisfactory with the agglomeration of CdS, which substantiates the observation of the phenomenon in the SEM. Moreover, although pristine CdS has the optimal specific surface area and porosity, the hydrogen production activity of the catalyst is not only related to the specific surface area of the sample, but also determined by factors such as the electron-hole pairs transfer rate. Table 1

3.2. Photo-piezoelectric hydrogen evolution properties

With the aim of investigating the photocatalytic performance of BFO/CdS heterojunction composites, the catalytic hydrogen evolution performance of catalysts with different composite ratios was tested and compared under visible-light irradiation in light-only, ultrasonic-only, and light coupled ultrasonic environments, respectively. $\text{Na}_2\text{S}/\text{Na}_2\text{SO}_3$ mixed solution was used as the sacrificial agent. The amount of hydrogen production versus time and the rate of hydrogen production for pristine BFO, pristine CdS and BFO/CdS heterojunction composites with different mass ratios under the effect of light-only conditions are shown in Fig. 6a and b. The amount of hydrogen evolution increases with time for all samples. The pristine BFO is limited by its excessive grain size that makes carriers transport in the bulk phase difficult, which in turn makes its hydrogen production performance unsatisfactory. Meanwhile, the pristine CdS phase accelerates the photogenerated carrier transport rate due to its thin-layer structure and high specific surface area, which not only benefits the adsorption of reactants and desorption of hydrogen, resulting in a hydrogen production rate of $0.54 \text{ mmol} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$ under visible light irradiation. The hydrogen production rates of all composites are higher than those of pristine BFO and CdS, with BC20 having a hydrogen evolution rate of $1.31 \text{ mmol} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$, a factor of 42.81 higher than that of pristine BFO. This not only confirms that its photocatalytic performance is related to the specific surface area as well as the separation efficiency of its photogenerated electron-hole pairs, but also demonstrates that the construction of the heterojunction formed between BFO and CdS effectively accelerates the separation of photogenerated carriers. To further confirm the piezoelectric response and piezocatalytic efficiency of each sample, ultrasound was introduced as an exogenous piezoelectric field for the piezocatalytic hydrogen production test, and the results are shown in Fig. 6c and d. In

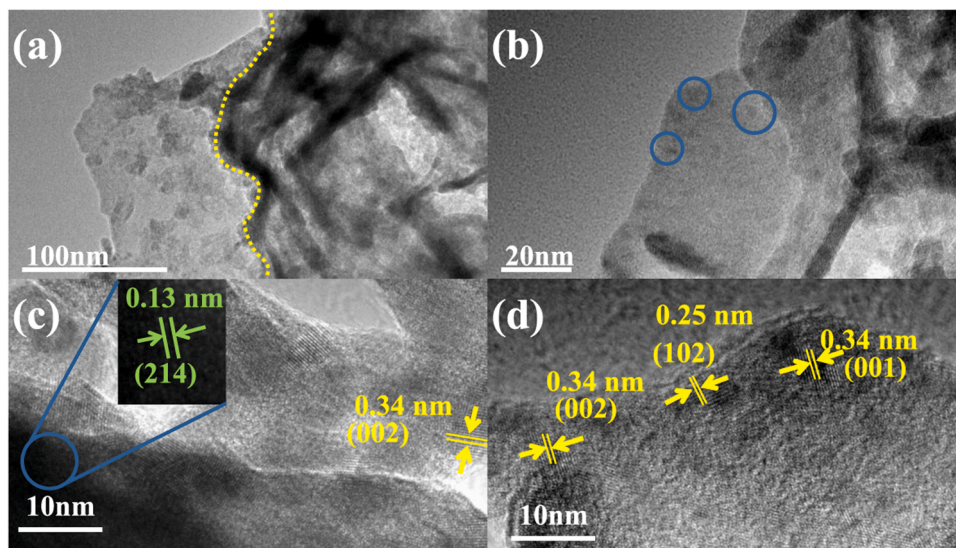


Fig. 4. TEM images of BC20 catalysts, (a) BFO/CdS composite morphology, (b) deposition of Pt on CdS nanosheets, (c) the lattice fringes of the two components and (d) lattice surface index of CdS nanosheets.

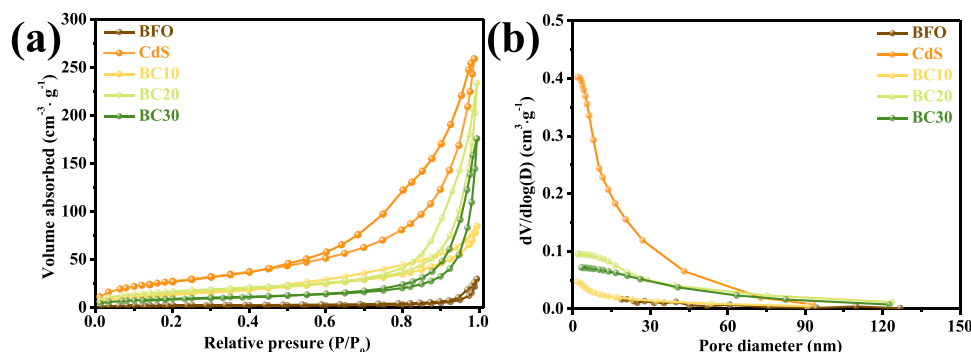


Fig. 5. (a) Nitrogen adsorption-desorption isotherms of all catalysts, (b) the pore size distribution curves of pristine BFO, CdS and BFO/CdS heterojunction composites.

Table 1

BET specific surface area and Pore Volume of pristine BFO, CdS and BFO/CdS heterojunction composites.

Sample	BET/m ² ·g ⁻¹	Pore Volume/cm ³ ·g ⁻¹
BFO	3.5423	0.061208
CdS	103.2473	0.321361
BC10	13.6362	0.088427
BC20	49.3286	0.234163
BC30	35.3446	0.141126

contrast to the photocatalysis, the difference in the rate of piezoelectric catalysis for hydrogen production is no longer significant, but the catalytic activity of the composite samples is much higher than that of the pure phase samples. Moreover, the spindle and ellipsoidal particle structure of the pristine BFO leads to a much lower frequency and degree of deformation than that of CdS with a 2D lamellar structure, which results in a relatively low piezoelectric catalytic activity. When visible light irradiation and ultrasound are applied to each catalyst sample simultaneously, the catalytic activity of each sample is much higher than the sum of photocatalysis and piezoelectric catalysis. This indicates that the coupling process of piezoelectric effect and photocatalysis forms a synergistic effect. In particular, the hydrogen production rate of 3.58 mmol·g⁻¹·h⁻¹ can be achieved by BC20 under the effect of multi-field coupling. With a view to expanding the range of applications of catalytic hydrogen production, the amount of hydrogen evolution from pure water splitting and the hydrogen evolution rate for each catalyst under coupled photopiezoelectric catalysis are given in Fig. 6g and h. The reaction of multi-field coupled catalytic pure water splitting is relatively slow in the initial stage due to the limitation of the four-electron reaction at the oxidation end, while there is a significant increase after 2 h and the hydrogen production rate of BC20 can reach 0.19 mmol·g⁻¹·h⁻¹.

Spontaneous polarization of catalyst initiated by an exogenous piezoelectric field leads to an internally polarized electric field inside the crystallites. The free charge within the grain migrates directionally towards the interface of the two components due to the effects of the internal electric field. The generation of another internal electric field between the two components in the cause of a heterojunction, which leads to further migration of photogenerated charges to the free charges in the bulk phase. In turn, the dual internal electric field constructed by multi-field coupling and heterogeneous structure can effectively inhibit the complexation of photogenerated carriers, which makes the composite catalyst equipped with the ability to pure water splitting. Additionally, the composite morphology of CdS nanosheets loaded on the BFO surface has a rougher surface and richer pore structure. Under the effect of ultrasound-driven reaction system flushing, the interfacial mass transfer efficiency of the catalyst is effectively accelerated, which in turn leads to a significant increase in the reactivity of photo-piezoelectric

catalytic hydrogen production.

The chemical stability of photocatalytic redox is a topic that cannot be bypassed for a long time. Since the stable perovskite crystal structure of BFO allows for good chemical stability, in turn, the easy photo-corrosive CdS was used for comparison with BC20. The Catalytic stability test of pristine CdS and BC20 under ultrasonic vibration and visible-light irradiation coupling conditions with sacrificial agent is shown in Fig. 6i. It can be found that BC20 has a good chemical stability, decaying only 10.1 % under 5 cycles of reaction. In contrast, the hydrogen production activity of pristine CdS decays by 31.7 % under 5 cycles, which is a significant decrease compared with BC20. This is due to the fact that the relative energy levels of the heterojunction interface determine the direction and efficiency of carrier separation. This is due to the fact that the relative energy levels of the heterojunction interface determine the direction and efficiency of carrier separation. In addition, the heterogeneous structure reduces the necessary charge migration distance and spatially separates the electron-hole pairs, effectively inhibiting bulk phase recombination. Whereas the pristine CdS is easily oxidised by photogenerated holes, leading to self-corrosion. The construction of Z-Scheme heterojunction can improve the stability of the composites by trapping photogenerated holes through BFO. Pure water splitting is thermodynamically more difficult and kinetically slower than in sacrificial agent containing systems. As shown in Fig. 6j, the hydrogen production cycle stability of pure water splitting by pristine CdS decays to 68.8 % after 5 cycles due to the excessive accumulation of photogenerated cavities on the surface [39,40], which much lower than that under the reaction system containing sacrificial agent (31.7 %), suggesting that the utilisation of the Na₂S/Na₂SO₃ sacrificial agent is able to lower the thermodynamic barrier to H₂O splitting as well as reduce the degree of photocorrosion of the catalyst. The XRD of BC20 before and after the reaction under the conditions of pure water and sacrificial agent is shown in Fig. S1, the position of the diffraction peaks of BC20 did not change after the reaction, and the intensity of the peaks is slightly weakened, which indicates that its crystallinity has decreased. This is also the reason why the activity of BC20 decreased after 5 cycles, but the overall catalytic activity remained high. There is a strong synergistic effect between sulfur compound photocatalysts and OH⁻ anions, and in the absence of hole sacrificers, OH⁻ can also greatly contribute to the surface kinetics of hole release and water decomposition [41]. The ·OH is the main product of the water oxidation half-reaction in the presence of OH⁻.

3.3. Optical and electrochemical properties

The composite photocatalysts show the higher photocatalytic activity than the pristine single-phase catalysts, which can be attributed to the construction of BFO/CdS heterostructures and the good interfacial contact between them to improve the photogenerated carriers separation efficiency. Thus, further studies on the optical and electrochemical

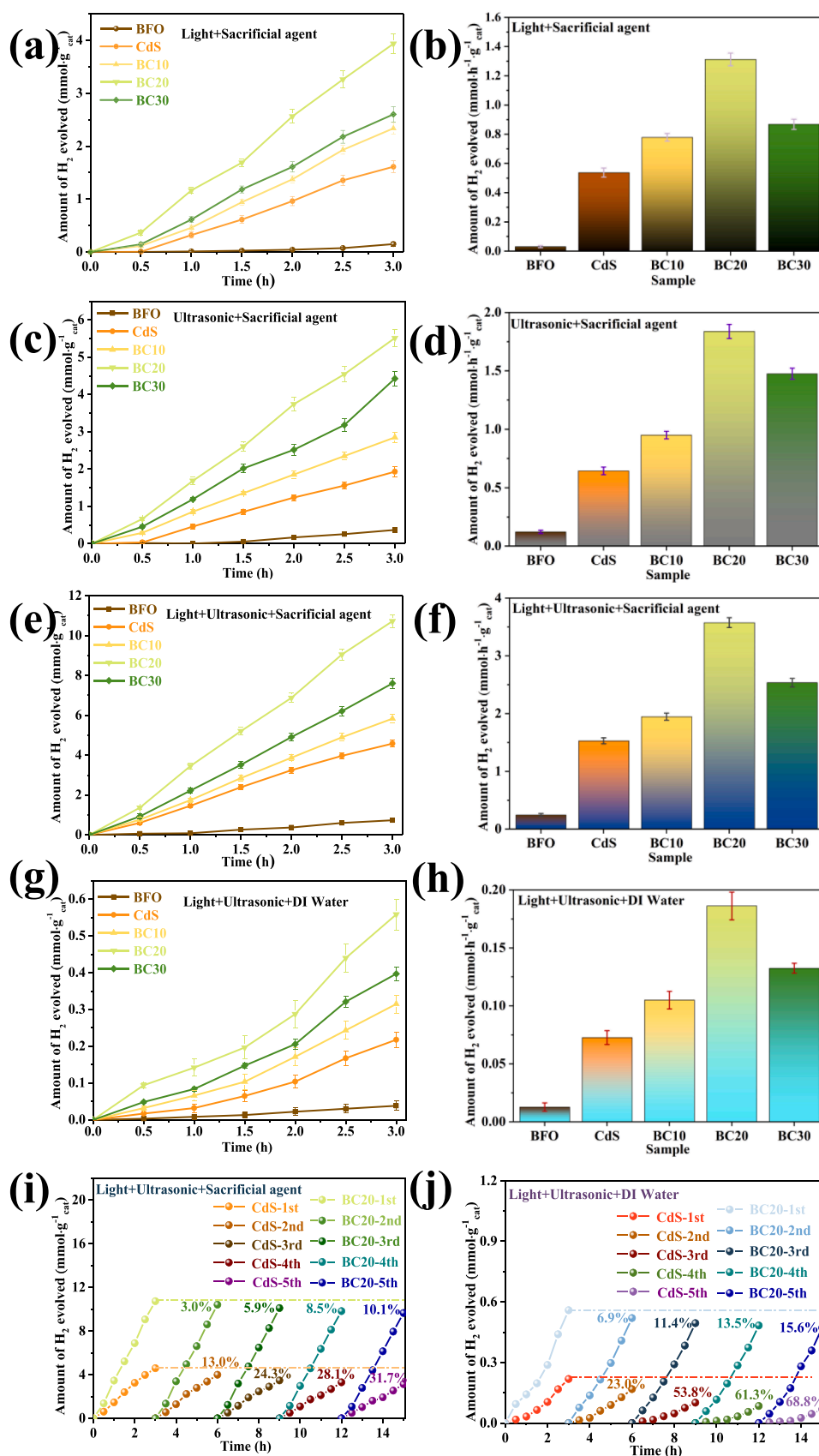


Fig. 6. The accumulated H_2 evolution overreaction time and the hydrogen production rate with $\text{Na}_2\text{S}/\text{Na}_2\text{SO}_3$ mixed solution as sacrificial reagent of pristine BFO, pristine CdS and BFO/CdS composite catalysts under (a,b) visible-light illumination, (c, d) ultrasonic vibration, (e, f) both ultrasonic vibration and visible-light irradiation and (g, h) pure water splitting under both ultrasonic vibration and visible-light irradiation condition. The Catalytic stability test of pristine CdS and BC20 under ultrasonic vibration and visible-light irradiation condition (i) with sacrificial agent and (j) without sacrificial agent.

properties of each catalyst were conducted to explore the enhancement mechanism. The light absorption capacity of the pristine BFO, CdS and heterojunction composite catalysts can be seen by UV-Vis absorbance spectra in Fig. 7a. All of the samples have excellent visible light absorption capacity, and the composites all show stronger light absorption performance than the pristine BFO or CdS in the wavelength range higher than 390 nm. The bandgap of BFO and CdS were estimated to be 2.13 eV and 2.51 eV respectively based on UV-Vis absorbance spectra combined with the Tauc equation as can be seen in Fig. 7b. The band structure of the two component samples constituting the heterojunction was further clarified by the Mott-Schottky test and the frequency was set to 800 and 1000 Hz. The slope of the Mott-Schottky curve for BFO can be found to be negative through Fig. 7c [42], while it is positive for CdS in Fig. 7d [43]. Therefore, the prepared BFO is a p-type semiconductor while CdS is an n-type semiconductor. From the bandgap values and the Mott-Schottky test, the location of the conduction and valence bands can also be estimated. The intersection of the different frequency linear parts of the Mott-Schottky diagram shows the flat band potential location, which corresponds to 1.42 and -1.26 eV vs Ag/AgCl for BFO and CdS, respectively. The flat band locations of the prepared BFO and CdS were transformed via the Nernst equation ($E_{RHE} = E_{Ag/AgCl} + 0.059 \times pH + E_{Ag/AgCl}^0$) to 2.02 eV and -0.66 eV for the standard hydrogen electrode potential. For the standard hydrogen electrode, the E_{VB} for p-type semiconductors is very close to the flat band potential value, while the

E_{CB} for n-type semiconductors is very close to the flat band potential value [44]. Therefore, the VB position of BFO is 2.02 eV and the CB position of CdS is -0.66 eV, and the corresponding conduction band or valence band positions can be deduced from $E_{CB} = E_{VB} - E_g$. Toward investigating the impact of heterogeneous structure construction of BC20 composites on the degree of carrier recombination, the transfer of photogenerated carriers was probed by the photoluminescence properties (PL) and the results are shown in Fig. 7e. The PL peak intensities of BFO/CdS are all decreased to different degrees compared with those of the pristine samples, indicating that the successful fabrication of heterojunction can be effective to inhibit the internal recombination of photogenerated carriers in the composite samples. It is noteworthy that the emission peaks of the heterojunction composites are significantly shifted to higher wavelengths compared with those of the pure BFOs, suggesting that the bandgap and the surface states of the heterojunction composites are changed. In addition, the heterojunction between 2D materials and nanoparticles may lead to tighter contacts between layered CdS and BFO as well as more efficient separation of electron-hole pairs. The photocurrent density and electrochemical impedance spectroscopy (EIS) tests were performed to verify the charge transfer and electrochemical properties of the heterojunction composites. As shown in Fig. 7f and g, the highest photocurrent intensity possessed by BC20 further proves that it has the highest separation efficiency for photogenerated carriers. Besides, the smaller diameter of the

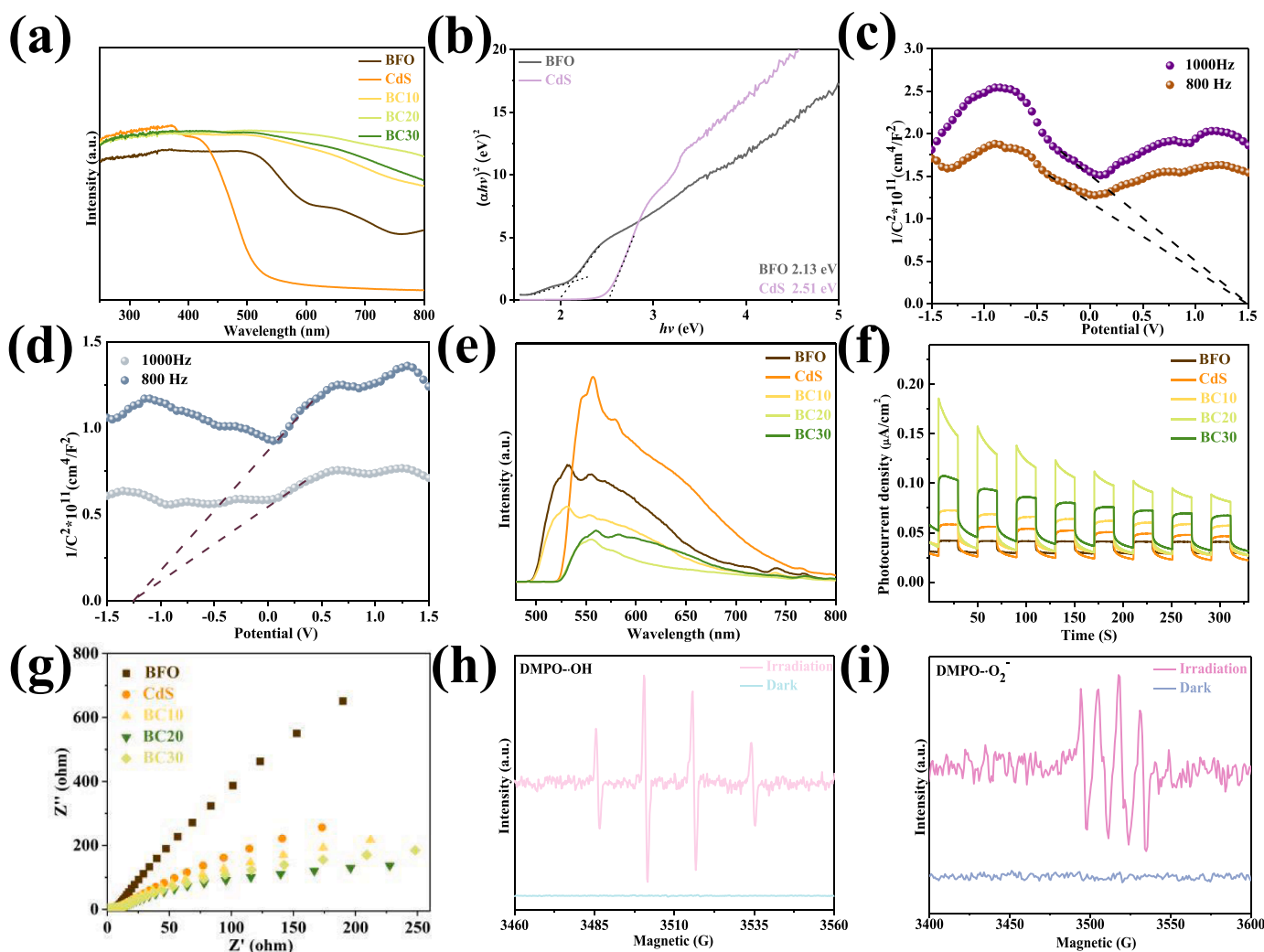


Fig. 7. (a) UV-Vis spectra and (b) Plots of $(\alpha h\nu)^{1/2}$ versus $h\nu$, (c) Mott-Schottky curves for BFO (d) Mott-Schottky curves for CdS, (e) PL spectra, (f) photocurrent density and (g) EIS plot for pristine BFO, CdS and BFO/CdS heterojunction composites. DMPO spin-trapping ESR spectra: (h) BC20- H_2O /DMPO, (i) BC20- CH_3OH /DMPO.

semicircular arc of the impedance curve in the electrochemical impedance spectra implies a faster charge transfer rate, which is conducive to enhanced photogenerated charge migration. The charge transfer resistance of the heterojunction composite samples is significantly reduced compared with that of pristine BFO and CdS by comparing the electrochemical impedance spectra. BC20 corresponds to the smallest diameter of the semicircular arcs of the impedance spectra indicating that the fastest charge transfer rate is achieved.

To more visually confirm whether this heterojunction follows a Z-scheme reaction route as well as to identify the reaction intermediate reactive species, it was investigated by electron paramagnetic resonance (ESR) spin trapping spectroscopy of DMPO- $\cdot\text{OH}$ measured in aqueous suspensions of the BC20 composite samples and DMPO- $\cdot\text{O}_2$ measured in methanol suspensions. The characteristic peaks of $\cdot\text{OH}$ and $\cdot\text{O}_2$ radicals were all detected in BC20 composite samples under xenon lamp irradiation. The CB position of CdS is more negative than the redox potential of $\text{O}_2/\cdot\text{O}_2$ (-0.33 eV vs RHE [45]). Therefore, it can provide electrons to produce $\cdot\text{O}_2$. However, the VB position of CdS is more negative than the oxidation potential of $\cdot\text{OH}/\text{OH}^\cdot$ (1.90 eV vs. RHE [46]), which prevents it from oxidizing $\text{OH}^\cdot$ to $\cdot\text{OH}$. On the contrary, in contrast, the VB and CB positions of BFO are more positive than $\cdot\text{OH}/\text{OH}^\cdot$ and $\text{O}_2/\cdot\text{O}_2$, which ensures its ability to oxidize $\text{OH}^\cdot$ to $\cdot\text{OH}$ but without $\cdot\text{O}_2$. Based on the above analysis, if the heterojunction of BFO/CdS is type-II, it is unable to produce $\cdot\text{OH}$ or $\cdot\text{O}_2$ radicals. The Z-scheme heterojunction composed of

BFO/CdS retains the more reducing CB of CdS and the more oxidizing VB of BFO are retained to obtain a more effective redox capacity.

Currently, photocatalysts do not have catalytic reaction continuity due to their high dependence on light illumination. The synergistic action of the external field that generates spontaneous polarization under piezoelectric effect can effectively compensate for the single dependence of the catalyst on light response. The catalytic hydrogen production activity has confirmed the synergistic effect between the photocatalytic reaction and the piezoelectric catalytic reaction. To further explore the piezoelectricity of the pristine BFO, pristine CdS and heterojunction composites more intuitively, the piezoelectric coefficient (d_{33}), piezoelectric current and piezoelectric voltage of each sample are shown in Fig. 8. The piezoelectric coefficients d_{33} of pristine BFO, CdS and composite samples with different ratios were measured to be 17.04, 20.24, 26.48, 35.2 and 29.5 $\text{pC}\cdot\text{N}^{-1}$, respectively, by the preparation of thin films for each sample. It is confirmed that each catalyst sample not only has a strong piezoelectric response, but also its piezoelectric coefficient coincides with the charge transport rate response in the electrochemical impedance spectroscopy plot, which also laterally verifies the reason for the difference in the reactivity of the piezocatalytic hydrogen production reaction. Further piezoelectric voltage and piezoelectric current tests were carried out on each catalyst sample. In the stressed state, the top and bottom surfaces of the catalyst films have different potentials and the potential difference was measured to be 0.15, 0.21, 0.32, 0.51,

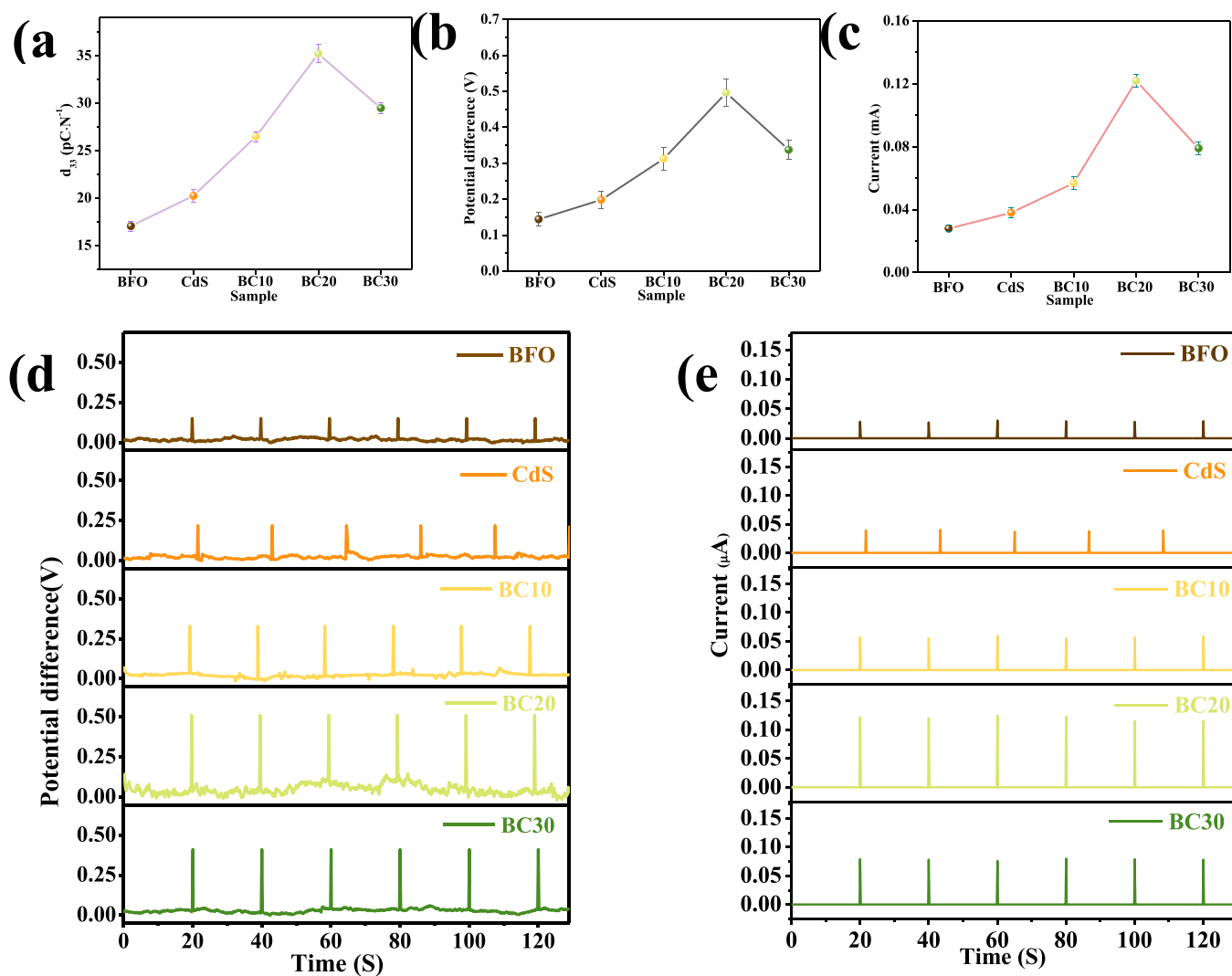


Fig. 8. (a) Measured d_{33} values of different samples, (b, d) measured electric potential difference, (c, e) output current of pristine BFO, CdS and heterojunction composite catalysts.

and 0.39 V for BFO, CdS and each of the heterojunction composites, respectively (Fig. 8b). The distance between the atomic layers decreases under applied stress, which in turn induces spontaneous polarisation and dipole moment generation. When the catalyst film is connected to an external circuit, the electrons and holes trapped at the surface are released and thus transient currents are observed. Similarly, the potential difference increases rapidly when the catalyst film is compressed and quickly returned to its original value as the external stress is removed. The measured output currents are shown in Fig. 8c, with 0.031, 0.039, 0.056, 0.078 and 0.125 μA for each catalyst sample under cyclic compression. It is evident that both BFO/CdS heterojunction composites exhibit higher piezoelectric generation performance than pristine BFO or CdS. As the center displacements of negative and positive ions (bound charges) in the single-phase crystal are lower than the Curie temperature, the spontaneous polarisation induced by external forces results in an internal electrostatic field within the BFO and CdS domains, which leads to the directional separation of free charges and photogenerated carriers in the catalysts with a piezoelectric response. It further leads to the concentration of free electrons on the positively polarised surface and the concentration of holes on the negatively polarised surface, which facilitates the conduct of redox reactions on the catalyst surface.

In order to further investigate the electron transfer pathways between the two components of BFO/CdS composites with different energy bands interleaved, the work function of the samples was calculated using density flood theory (DFT). The calculation of the work function (W) is important for determining the energy band edge potentials and thus analysing the redox properties. The formula is expressed as $W = E_{\text{vac}} - E_{\text{F}}$, where E_{vac} denotes the vacuum energy level of the material

and E_{F} denotes the Fermi energy level of the material. As shown in Fig. 9a and b, the work functions of BFO and CdS are 2.67 and 4.65 eV, respectively, and thus the Fermi energy level of BFO is higher than that of CdS. When the two single-phase materials contact at the interface to form a heterojunction, the electrons will be transferred from BFO to CdS. At the same time, an internal electric field is formed from the surface of CdS to the surface of BFO. Until the Fermi energy level reaches the equilibrium value, the work function of the BFO/CdS composite is 4.23 eV, which results in the formation of a Z-scheme charge transfer pathway, as shown in Fig. 9c.

3.4. The proposed mechanism

The photogenerated electrons are unstable in the excited state and can easily return to the ground state to compound with the holes. Moreover, due to the influence of impurities in photocatalytic materials on the trapping of carriers, it is easy to form a carrier complex center near the impurities, so that single photocatalytic materials have low intrinsic carrier separation efficiency, which seriously affects their photocatalytic efficiency. Heterojunction composites are designed to achieve the spontaneous movement of photogenerated carriers from high to low energy levels through the differences in the band structures of different semiconductors and the formation of a spontaneous internal electric field at the composite interface through the potential energy difference, which improves the separation efficiency of electron-hole pairs in a photocatalyst. The formation of Z-Scheme heterojunctions was confirmed by band structure testing and ESR spectroscopy, combined with the electron flow direction confirmed in TEM. The charge transfer schematic of the BFO/CdS heterojunction composite is shown in

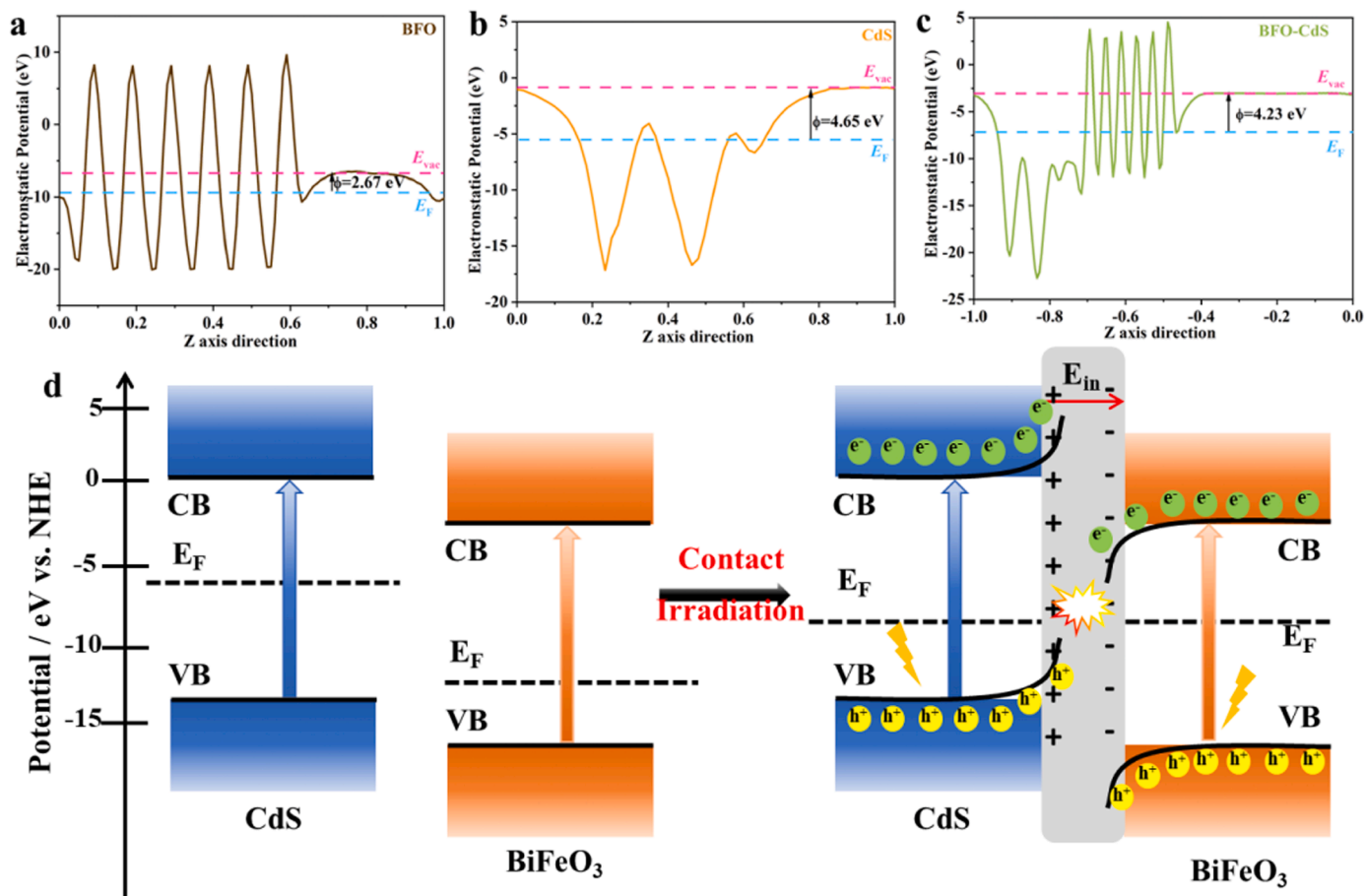


Fig. 9. (a-c) Electrostatic potentials of BFO, CdS, and BFO/CdS surface by work functions. (d) (Left) The electronic energy band structure diagram of BFO and CdS, (right) the formed Z-scheme heterojunction between BFO and CdS.

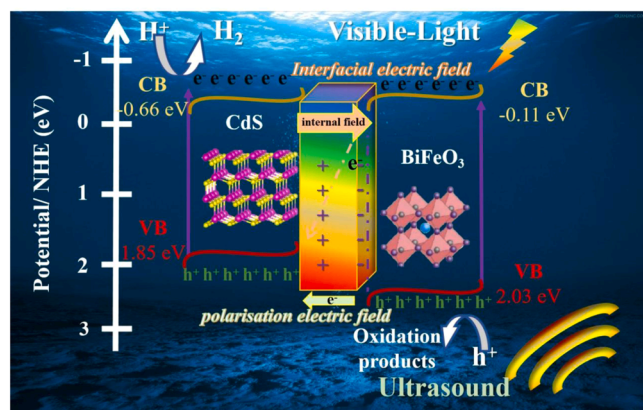


Fig. 10. The schematic of charge separation of Z-Scheme heterojunction BFO/CdS under the coupling of piezoelectric effect and light irradiation.

Fig. 10. Since the EF of CdS is higher than that of BFO, and thus when CdS is in contact with BFO, electrons spontaneously migrate from the CdS surface to the BFO surface, leading to a new EF equilibrium. In the equilibrium state, positive charges accumulate at the CdS interface and the energy band bends upwards due to electron efflux. On the contrary, due to the inflow of electrons, negative charges accumulate at the interface of the BFO, resulting in downward bending of the energy bands. Meanwhile, the redistribution of charges at the CdS/BFO interface generates an inbuilt electric field, which facilitates the binding of electrons in the CB of the BFO and holes in the VB of the CdS. Under visible light irradiation, the electrons in the CB of BFO were driven by the internal electric field to complex with the holes in the VB of CdS. Under irradiation, the constructed Z-scheme heterojunction can produce a high degree of spatial separation of charge carriers and redox sites on the composite photocatalyst. In addition, the CdS nanosheets as mass transfer carriers with ultrathin structure can desorb the generated hydrogen molecules from the photocatalyst surface in a timely manner, which is important for improving the photocatalytic reaction rate, avoiding the reverse reaction, and releasing the active sites.

Since BFO and CdS are not only photocatalysts but also proved to have excellent piezoelectric response, there is both an interfacial electric field formed by the Z-Scheme heterojunction and a polarisation electric field between the two components. The catalyst with an asymmetric centre is subjected to an external piezoelectric field which generates a polarised electric field inside the crystal. As the saturated piezoelectrically polarised electric field of the piezoelectric semiconductor is completely cancelled out by the surface shielding charge, the saturated polarisation strength of the piezoelectric semiconductor reaches a maximum value, the net polarisation strength becomes zero and the surface is not chargeable. With the reduction of the saturation piezoelectric polarisation strength, the piezoelectric polarisation field decreases and the excess shielding charge located on the surface of the piezoelectric semiconductor is released, which in turn drives the redox reaction with the reactants in effective contact with it to occur with the loss and gain of electrons. If the piezoelectric polarisation strength is increased under external stress, the piezoelectric semiconductor surface will adsorb spatially anisotropic free charges to compensate for the increasing piezoelectric polarisation field, and the piezoelectric semiconductor surface in effective contact with the reactants are also due to the reactive medium or the piezoelectric semiconductor charge exchange and the redox properties. In this way, the free charge density on the surface of the piezoelectric semiconductor decreases or increases continuously due to the periodic decrease or increase of the piezoelectric polarisation intensity, which causes the piezoelectric semiconductor to exhibit catalytic reaction activity.

When light and external stress act simultaneously on single components of BFO or CdS catalysts, the piezoelectric potential as a strong

driving force can promote the separation of photogenerated carriers from free charges within the bulk phase in piezoelectric photocatalysts. The redistribution of polarisation charges can change the interface and bulk phase energy distribution, thus affecting the carrier migration inside the material and at the interface. The polarity generated through the piezoelectric potential can modulate the effective Schottky barrier height near the contact surface and further enhance the separation through the Z-Scheme heterojunction. Specifically, the piezoelectrically polarised charge from the piezoelectric will induce energy band bending at the interface, which attracts photogenerated carriers with opposite charges. Consequently, the photogenerated electrons in the BFO CB are rapidly transferred through the contact interface, and furthermore, the hydrogen generation reaction is accelerated due to the enhancement of the ferroelectric polarisation of the CdS and the BFO which makes the CB of the CdS sufficiently charged with electrons.

4. Conclusions

In this thesis, BFO/CdS heterojunction composite catalysts are constructed based mainly on the perovskite oxide BFO, and the multiferroic nature of the BFO-based materials is optimised by constructing it with CdS nanosheets. Meanwhile, the composite materials have been built to strengthen the visible light absorption ability and the charge transfer rate between the two components. The constructed Z-Scheme heterojunction reduces the recombination of photogenerated carriers and enhances the interfacial charge transportation as well as the adsorption and desorption rates of the reaction medium, as corroborated by band structure calculations, PL spectroscopy, radical trapping experiments, and catalytic hydrogen production performance. In this case, the visible light catalysis hydrogen production rate of BC20 reached $1.31 \text{ mmol} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$, which is 42.81 times higher than that of the pristine BFO.

Further, it is verified that both BFO and CdS have not only a visible light response but also favorable piezoelectricity. The migration rates of bulk-phase free charges and photogenerated carriers are enhanced by double internal electric field coupling of the Z-Scheme heterojunction and the piezoelectric polarisation electric field. The photo-piezoelectric catalytic efficiency of the composites is significantly higher than the sum of hydrogen production from the piezoelectric and photocatalytic reactions. It is further revealed that photogenerated electrons and holes are more prone to migrate in the direction of the piezoelectric polarisation field strength of BFO/CdS induction. The separation and transfer rates of photogenerated carriers are significantly enhanced by the synergy of interfacial transport induced by Z-Scheme heterojunction and polarisation field triggered by piezoelectric effect. Meanwhile, the retained more negative CB of CdS and the more positive VB of BFO further optimised the redox capacity of the composite catalysts. The experimental results show that the BC20 composite achieves a hydrogen production rate of $3.58 \text{ mmol} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$ by photo-piezoelectric catalysis in a sacrificial agent environment, and also possesses the ability to deionised water splitting. This study provides a promising approach to further enhance the charge migration method of perovskite under multifield coupling reaction conditions.

CRedit authorship contribution statement

Yutian Li: Data curation. Zehua Zhang: Conceptualization. Jiaoyang Qu: Formal analysis, Data curation, Conceptualization. Meidan Que: Funding acquisition, Formal analysis. Die Zhou: Data curation. Kui Nie: Formal analysis. Cheng Wang: Conceptualization. Qing Ji: Data curation. Zhao Li: Data curation. Rui Li: Investigation. Yongjie Cao: Data curation. Long Zhang: Conceptualization. Xinyu Ma: Data curation, Conceptualization.

Authors' statement

Xinyu Ma, Rui Li, Zhao Li, Long Zhang and Jiaoyang Qu contributed to the initiating idea and designed the experiments. Yongjie Cao, Kui Nie, Die Zhou, Qing Ji, Cheng Wang, Zehua Zhang, Yutian Li, Meidan Que conducted the piezoelectric property measurements. Xinyu Ma and Jiaoyang Qu analyzed the data and wrote the manuscript. All authors commented on the final manuscript and had given approval to the final version.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was financially supported by the Guizhou Provincial Science and Technology Project (NO. 2023–134), and the Collaborative Innovation Center Project of Shaanxi Provincial Department of Education (NO. 23JY045).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jece.2024.115049](https://doi.org/10.1016/j.jece.2024.115049).

Data availability

Data will be made available on request.

References

- [1] F.Z. Su, S.C. Mathew, G. Lipner, X.Z. Fu, M. Antonietti, S. Blechert, X.C. Wang, mpg-C₃N₄-Catalyzed selective oxidation of alcohols using O₂ and visible light, *J. Am. Chem. Soc.* 132 (2010) 16299–16301.
- [2] J. Zhang, Y. Zhang, L. Li, W. Yan, A. H. Wang, W. Mao, Synergizing the internal electric field and ferroelectric polarization of the BiFeO₃/ZnIn₂S₄ Z-scheme heterojunction for photocatalytic overall water splitting, *J. Mater. Chem. A* 11 (2023) 434.
- [3] J. Zhang, L. Wang, X. Liu, X. Li, W. Huang, High-performance CdS-ZnS core-shell nanorod array photoelectrode for photoelectrochemical hydrogen generation, *J. Mater. Chem. A* 3 (2015) 535.
- [4] S. Horikoshi, F. Hojo, H. Hidaka, N. Serpone, Environmental remediation by an integrated microwave/UV illumination technique. 8. fate of carboxylic acids, aldehydes, alkoxycarbonyl and phenolic substrates in a microwave radiation field in the presence of TiO₂ particles under UV irradiation, *Environ. Sci. Technol.* 38 (2004) 2198–2208.
- [5] S. Li, Z.C. Zhao, J.Z. Zhao, Z.T. Zhang, X. Li, J.M. Zhang, Recent advances of ferro-, piezo-, and pyroelectric nanomaterials for catalytic applications, *ACS Appl. Energy Mater.* 3 (2020) 1063–1079.
- [6] P.C. Sevinc, B. Dhital, V.G. Rao, Y.M. Wang, H.P. Lu, Probing electric field effect on covalent interactions at a molecule-semiconductor interface, *J. Am. Chem. Soc.* 138 (2016) 1536–1542.
- [7] S. Dunn, P.M. Jones, D.E. Gallardo, Photochemical growth of silver nanoparticles on c⁻ and c⁺ domains on lead zirconate titanate thin films, *J. Am. Chem. Soc.* 129 (2007) 8724–8728.
- [8] C. Shu, A. Noble, V.K. Aggarwal, Photoredox-catalyzed cyclobutane synthesis by a deboronative radical addition-polar cyclization cascade, *Angew. Chem. Int. Ed.* 58 (2019) 3870–3874.
- [9] D. Vanderbilt, Principles and applications of ferroelectrics and related materials, *Mater. Res. Bull.* 11 (2002) 450–451.
- [10] J. Li, L.J. Cai, J. Shang, Y. Yu, L.Z. Zhang, Giant enhancement of internal electric field boosting bulk charge separation for photocatalysis, *Adv. Mater.* 28 (2016) 4059–4064.
- [11] Z.Z. Lou, B.B. Huang, Z.Y. Wang, X.C. Ma, R. Zhang, X.Y. Zhang, X.Y. Qin, Y. Dai, M.H. Whangbo, Ag₆Si₂O₇: a silicate photocatalyst for the visible region, *Chem. Mater.* 26 (2014) 3873–3875.
- [12] Y. Yu, W. Yan, X.F. Wang, P. Li, W.Y. Gao, H.H. Zou, S.M. Wu, K.J. Ding, Surface engineering for extremely enhanced charge separation and photocatalytic hydrogen evolution on g-C₃N₄, *Adv. Mater.* 30 (2018) 1705060.
- [13] X. Wang, Y. Zhang, C. Zhou, D. Huo, R. Zhang, L. Wang, Hydroxyl-regulated BiOI nanosheets with a highly positive valence band maximum for improved visible-light photocatalytic performance, *Appl. Catal. B Environ. Energy* 268 (2020) 118390.
- [14] J. Wang, D. Jin, H. Mei, Q. Lin, R. Zhang, X. Wang, In situ construction of BiO (ClBr)_{(1-x)/2(1-x)} solid solution with appropriate surface iodine vacancies for synergistically boosting visible-light photo-oxidation capability, *Inorg. Chem.* 62 (2023) 1539–1548.
- [15] Y.L. Liu, J.M. Wu, Synergistically catalytic activities of BiFeO₃/TiO₂ core-shell nanocomposites for degradation of organic dye molecule through piezophototronic effect, *Nano Energy* 56 (2019) 74–81.
- [16] S.Y. Lan, B.H. Jing, C. Yu, D.M. Yan, Z. Li, Z.M. Ao, M.S. Zhu, Protrudent iron single-atom accelerated interfacial piezoelectric polarization for self-powered water motion triggered fenton-like reaction, *Small* 18 (2022) 2105279.
- [17] K.S. Hong, H.F. Xu, H. Konishi, X.C. Li, Direct water splitting through vibrating piezoelectric microfibers in water, *J. Phys. Chem. Lett.* 1 (2010) 997–1002.
- [18] M.B. Starr, J. Shi, X.D. Wang, Piezopotential-driven redox reactions at the surface of piezoelectric materials, *Angew. Chem. Int. Ed.* 51 (2012) 5962–5966.
- [19] K.S. Hong, H.F. Xu, H. Konishi, X.C. Li, Piezoelectrochemical effect: a new mechanism for azo dye decolorization in aqueous solution through vibrating piezoelectric microfibers, *J. Phys. Chem. C* 116 (2012) 13045–13051.
- [20] S.Y. Lan, B.H. Jing, C. Yu, D.M. Yan, Z. Li, Z.M. Ao, M.S. Zhu, Protrudent iron single-atom accelerated interfacial piezoelectric polarization for self-powered water motion triggered fenton-like reaction, *Small* 18 (2022) 2105279.
- [21] K. Kubota, Y.D. Pang, A. Miura, H. Ito, Redox reactions of small organic molecules using ball milling and piezoelectric materials, *Science* 366 (2019) 1500.
- [22] P. Zhu, Y. Chen, J.L. Shi, Piezocatalytic tumor therapy by ultrasound-triggered and BaTiO₃-mediated piezoelectricity, *Adv. Mater.* 32 (2020) 2001976.
- [23] Y. Wang, X.R. Wen, Y.M. Jia, M. Huang, F.F. Wang, X.H. Zhang, Y.Y. Bai, G. L. Yuan, Y.J. Wang, Piezo-catalysis for nondestructive tooth whitening, *Nat. Commun.* 11 (2020) 1328.
- [24] T. Fujii, S.I. Kawasaki, A. Suzuki, T. Adschiri, High-speed morphology control of boehmite nanoparticles by supercritical hydrothermal treatment with carboxylic acids, *Cryst. Growth Des.* 16 (2016) 1996–2001.
- [25] Q. Lin, C. Luo, D. Jin, L. Zhou, R. Zhang, X. Wang, Precise tuning of bilayer ultrasmall MoS₂ featuring inhibition of carrier recombination and fast surface chemistry for green H₂ evolution, *Small* 20 (2024) 2305888.
- [26] Q. Lin, S. Liang, J. Wang, R. Zhang, G. Liu, X. Wang, Hierarchically periodic macroporous CdS-ZnO heterojunctions with multiple quantum well-like band alignments for efficient photocatalytic hydrogen evolution without a cocatalyst, *ACS Sustain. Chem. Eng.* 11 (2023) 3093–3102.
- [27] X. Wang, L. Yin, G. Liu, Light irradiation-assisted synthesis of ZnO-CdS/ reduced graphene oxide heterostructured sheets for efficient photocatalytic H₂ evolution, *Chem. Commun.* 50 (2014) 3460.
- [28] J. Jin, J.G. Yu, D.P. Guo, C. Cui, W.K. Ho, A hierarchical Z-scheme CdS-WO₃ photocatalyst with enhanced CO₂ reduction activity, *Small* 11 (2015) 5262–5271.
- [29] G. Yu, L.P. Liu, C. Zhao, S.Q. Guo, H.B. Meng, Y.F. Zhu, X.M. Zhang, Enhanced internal electric field of CdS/NU-M Z-scheme heterojunction for efficient photocatalytic hydrogen evolution, *Chem. Eng. J.* 494 (2024) 153099.
- [30] R. Wu, F.Y. Chen, Y.B. Tang, Y.H. Song, W.Q. Sun, J.Y. Yang, Construction of Z-Scheme BiFeO₃/Bi₂O₄ heterojunction for significantly boosted photocatalytic degradation of tetracycline under visible irradiation, *Mater. Sci. Semicond. Proc.* 180 (2024) 108519.
- [31] S. Han, C. Sung Kim, Weak ferromagnetic behavior of BiFeO₃ at low temperature, *J. Appl. Phys.* 113 (2013) 17D921.
- [32] K. Chakrabarti, K. Das, B. Sarkar, S. Ghosh, Enhanced magnetic and dielectric properties of Eu and Co co-doped BiFeO₃ nanoparticles, *Appl. Phys. Lett.* 101 (2012) 042401.
- [33] P.S.V. Mocherla, C. Karthik, R. Ubig, M.S.R. Rao, C. Sudakar, Tunable bandgap in BiFeO₃ nanoparticles: the role of microstrain and oxygen defects, *Appl. Phys. Lett.* 103 (2013) 022910.
- [34] S.M.D. Lima, M.A. Pena, J.L.G. Fierro, J.M. Assaf, La_{1-x}Ca_xNiO₃ perovskite oxides: characterization and catalytic reactivity in dry reforming of methane, *Catal. Lett.* 124 (2008) 195–203.
- [35] X.H. Zhang, C.L. Pei, X. Chang, S. Chen, R. Liu, Z.J. Zhao, R.T. Mu, J.L. Gong, FeO₆ octahedral distortion activates lattice oxygen in perovskite ferrite for methane partial oxidation coupled with CO₂ splitting, *J. Am. Chem. Soc.* 142 (2020) 11540–11549.
- [36] X. Xu, H.Q. Wang, M. Fronzi, X.F. Wang, L. Bi, E. Traversa, Tailoring cations in a perovskite cathode for proton-conducting solid oxide fuel cells with high performance, *J. Mater. Chem. A* 7 (2019) 20624–20632.
- [37] Q.Q. Ji, X. Xu, X.H. Liu, L. Bei, Improvement of the catalytic properties of porous lanthanum manganite for the oxygen reduction reaction by partial substitution of strontium for lanthanum, *Electrochem. Commun.* 124 (2021) 106964.
- [38] C. Rudaz, R. Courson, L. Bonnet, S. Calas, H. Sallee, T. Budtova, Aeropectin: fully biomass-based mechanically strong and thermal superinsulating aerogel, *Biomacromolecules* 15 (2014) 2188–2195.
- [39] L. Qin, X. Xin, R. Wang, H. Lv, G. Yang, Rational design of bromine-modified Ir(III) photosensitizer for photocatalytic hydrogen generation, *J. Catal.* 417 (2023) 89–97.
- [40] S. Liu, Y. Yang, W. Xiao, S. Xia, C. Jin, W. Wang, S. Li, M. Zhong, S. Wang, C. Chen, Metal-organic frameworks derived porous MoS₂/CdS heterostructure for efficient photocatalytic performance towards hydrogen evolution and organic pollutants, *Int. J. Hydrog. Energ.* 48 (2023) 32729–32738.
- [41] W.H. Xue, X. Bai, J.Z. Tian, X.Y. Ma, J. Fan, E.Z. Liu, Enhanced photocatalytic H₂ evolution on ultrathin Cd_{0.5}Zn_{0.5}S nanosheets without a hole scavenger: combined analysis of surface reaction kinetics and energy-level alignment, *Chem. Eng. J.* 428 (2022) 132608.

- [42] S. Das, P. Fourmont, D. Benetti, S.G. Cloutier, R. Nechache, Z.M. Wang, F. Rosei, High performance BiFeO₃ ferroelectric nanostructured photocathodes, *J. Chem. Phys.* 153 (2020) 084705.
- [43] S. Smairi, B. Hartiti, Y. Arba, H. Labrim, S. Fadili, M. Tahri, A. Belfhailli, P. Thevenin, Elaboration and characterization of cadmium sulfide (CdS) thin films prepared by chemical bath deposition (CBD), *Mater. Today. Proc.* 66 (2022) 112–115.
- [44] J.H. Zheng, L. Zhang, Incorporation of CoO nanoparticles in 3D marigold flowerlike hierarchical architecture MnCo₂O₄ for highly boosting solar light photooxidation and reduction ability, *Appl. Catal. B Environ. Energy* 237 (2018) 1–8.
- [45] C.X. Li, H.N. Che, C.B. Liu, G.B. Che, P.A. Charpentier, W.Z. Xu, X.Y. Wang, L. H. Liu, Facile fabrication of g-C₃N₄ QDs/BiVO₄ Z-scheme heterojunction towards enhancing photodegradation activity under visible light, *J. Taiwan Inst. Chem. E* 95 (2019) 669–681.
- [46] C.C. Chen, W.H. Ma, J.C. Zhao, Semiconductor-mediated photodegradation of pollutants under visible-light irradiation, *Chem. Soc. Rev.* 39 (2010) 4206–4219.